
Persona Setting Pitfall: Persistent Outgroup Biases in Large Language Models Arising from Social Identity Adoption*

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ABSTRACT

Drawing parallels between human cognition and artificial intelligence, we explored how large language models (LLMs) internalize identities imposed by targeted prompts. Informed by Social Identity Theory, these identity assignments lead LLMs to distinguish between “we” (the ingroup) and “they” (the outgroup). This self-categorization generates both ingroup favoritism and outgroup bias. Nonetheless, existing literature has predominantly focused on ingroup favoritism, often overlooking outgroup bias, which is a fundamental source of intergroup prejudice and discrimination. Our experiment addresses this gap by demonstrating that outgroup bias manifests as strongly as ingroup favoritism. Furthermore, we successfully mitigated the inherent pro-liberal, anti-conservative bias in LLMs by guiding them to adopt the perspectives of the initially disfavored group. These results were replicated in the context of gender bias. Our findings highlight the potential to develop more equitable and balanced language models.

Keywords Large Language Model · Intergroup Bias · Social Psychology

1 Introduction

Large language models (LLMs) reproduce social biases, potentially causing negative impacts on users and societies. For instance, studies have found a pro-liberal bias on the political spectrum of generated responses by LLMs [1, 2, 3]. Gender stereotypes are reinforced in recommendation letters produced by LLMs, where female students are often described with likable personality traits and male students are highlighted for their leadership competence [4]. Racial biases are also present, particularly against African Americans, and can lead to less accurate predictions in health domains [5].

These biases have been extensively discussed within the framework of the parallels between human cognition and artificial intelligence [6, 7, 8]. The personalization of LLM outputs has further enriched this discourse [9, 10, 11]. Specifically, Simmons [12] found that when LLMs were assigned a distinct political identity through user prompting, their responses tended to align closely with the provided liberal or conservative cues. Feng et al. [13] further noted that LLMs designed with a specific viewpoint, such as right-leaning or left-leaning, performed better in detecting misinformation and hate speech directed against the group they were aligned with. These examples effectively demonstrate the capability of LLMs to emulate human cognitive biases.

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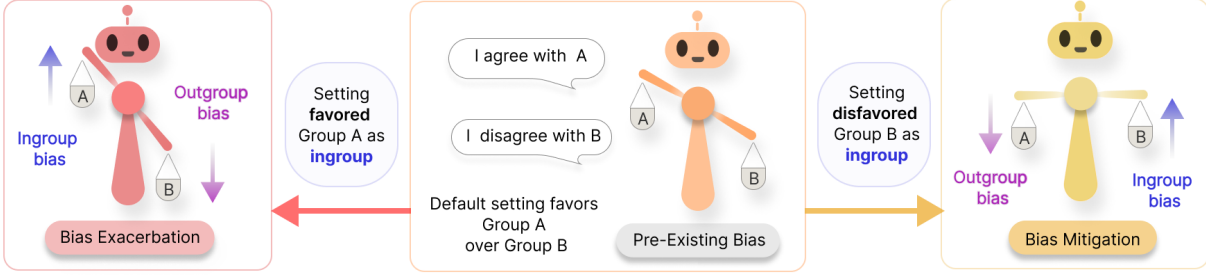


Figure 1: LLMs exhibit ingroup bias by aligning their values with the social identities present in prompts, while displaying outgroup bias by rejecting values associated with outgroup identities. Assigning a favored group identity exacerbates the pre-existing bias, whereas configuring a disfavored group identity can significantly mitigate it.

Social identity theory (SIT) [14, 15, 16] indicates a significant oversight in current machine psychology research. SIT posits that when individuals perceive themselves as part of a group, such as Democrats (vs. Republicans), women (vs. men), and Blacks (vs. Whites), they engage with the most fundamental distinction between “we,” the ingroup, and “they,” the outgroup. This social categorization drives the motivation to evaluate the ingroup positively (i.e., ingroup bias) while viewing the outgroup negatively (i.e., outgroup bias), often leading to ingroup favoritism and outgroup derogation (see [17] for an extensive review). Outgroup bias, rather than preferential treatment of the ingroup, constitutes the core of intergroup prejudice, animus, and social exclusion. Addressing both ingroup and outgroup biases is therefore crucial for understanding and mitigating biases within LLMs toward various social groups.

Nonetheless, existing research has largely overlooked outgroup bias, presenting an opportunity to develop fairer and more balanced models. Informed by social identity theory, we hypothesized in this research that assigning a particular social identity to LLMs through prompting would amplify their support for ingroup attributes while heightening their opposition to outgroup attributes. In scenarios devoid of initial preferences, this process would instill intergroup bias. Conversely, when an LLM exhibits pre-existing biases, prompting it to adopt the perspective of the “disfavored” group will help mitigate these biases. The targeted prompt is expected to increase support for the disfavored group (now perceived as the ingroup) while diminishing support for the initially favored group (now perceived as the outgroup). This approach may neutralize the inherent bias, if not reverse its direction (see research design in Figure 1).

We first tested these hypotheses within a political context. Using the measures from the political compass study [13], we mapped the political bias within LLMs, including GPT-4o. Consistent with extant literature, LLMs showed a marked preference for liberal values over conservative ones. Subsequently, we measured both ingroup and outgroup biases by assigning either a Democratic or Republican identity through specific prompts. Our results found that outgroup bias was as pronounced as, if not stronger than, the ingroup bias. Furthermore, the interplay of these ingroup and outgroup biases alleviated the initial inclinations, leading to a more balanced perspective. We show robustness of this methodology by repeating analysis in the context of gender bias.

Our study advances current literature in four key aspects. First, we extend existing research by quantifying the social biases in LLMs using psychological batteries with Likert scales. Despite its limitations, the method is cost-effective compared to those involving human annotation and open-ended questions. Second, we introduce the concept of ingroup and outgroup biases, which can be applied across various group dynamics, along with formulas to compute these biases. By doing so, we broaden the scope of comprehending and analyzing social biases in LLMs. Third, our findings indicate that when LLMs present an inherent bias against a certain social group, our hypothesized strategy of ‘setting the disfavored group as the ingroup through prompting’ could instantly adjust the models’ perspective. We discuss the potential applications of these findings and limitations of perspective recalibration methods within the machine psychology literature.

Lastly, contrary to our expectations, GPT-4o displayed an ungrounded preference for women over men in a wide range of survey questions posed, showing strong opposition to hostile sexism against women. When the LLM profile was set to associate men as the ingroup, GPT-4o also expressed strong anti-sexism towards men. Additionally, prompting a Republican identity produced a similar bias correction effect. This finding supports the concept of *algorithmic fidelity* [9], which is analogous to partisan sorting where social identities including racial, gender, and religious identities

becoming increasingly aligned with partisan identity [18, 19]. Our findings highlight that perspective value alignment methods can be applied broadly, not just in case-specific scenarios.

2 Related Work

2.1 Humanlike Biases in LLMs

LLMs not only demonstrate human-level capabilities [6, 20, 8, 21], but they also exhibit biases similar to those found in humans [22, 23]. For instance, LLMs generate content that reflects humanlike preferences [7]. However, since many LLMs are primarily trained on biased datasets, resulting in misalignments such as stereotypes related to gender and race, as well as political bias [13].

Humanlike biases manifest in different domains. LLMs exhibit political bias and generate content that disproportionately favors certain political viewpoints or ideologies [24, 10]. Research indicates that LLM responses to survey questions frequently exhibit a bias towards liberal perspectives [3, 1, 13]. Additionally, LLMs that align with Democratic Party viewpoints often depict Republican values and leaders unfavorably [2, 25]. Gender stereotypes are also present in LLM output. Language models stereotypically associate gender pronouns with different occupations [26]. They can also generate gender biased news content [27]. ChatGPT, specifically, has been shown to produce gender-biased responses [28] and to write recommendation letters that reinforce gender stereotypes [4].

Understanding the values reflected by LLMs necessitates examining both default data representations and persona-steered content generation. Consequently, some research has focused on quantifying the steerability of LLMs by investigating how these systems adopt personas and generate content accordingly [29, 30]. The balance between improved personalization through persona-steered generation and the reinforcement of biases, such as confirmation bias, has been a subject of discussion [11]. However, whether imposing personas to better align LLM responses with one social group will result in bias-level changes for other social groups remains insufficiently explored.

2.2 Bias Evaluations and Mitigations

Evaluating and mitigating biases in LLMs is essential for ensuring fair applications [31]. To assess stereotypes and biases, several studies have adopted methods from social science and psychology, including replicating human-designed surveys [9] and using crowdsourcing for bias annotation [32]. These evaluations are critical steps toward understanding and reducing biases in LLM responses [33, 22]. To correct LLM biases, recent efforts have focused on debiasing models through the cleaning of training data [13] and alignment post-processing [34, 24]. Although these methods significantly improve model behavior, they are computationally intensive. Bias detection and estimation in LLMs are often conducted in a zero-shot setting, with techniques such as adding debiasing instructions to downstream task prompts [35, 36, 37]. However, these LLM bias evaluation and correction methods do not fully account for intergroup dynamics, which is a crucial aspect of the bias sources.

2.3 Ingroup and Outgroup Biases: A Social Identity Approach

Social identity and group membership are fundamental concepts for humans to understand intergroup relations [38, 39, 40]. Social identity theory (SIT [14, 15, 16]) posits that when individuals perceive themselves as part of a group, they engage with the most fundamental distinction between “we,” the ingroup, and “they,” the outgroup. This social categorization can be based on various factors like gender, race, political affiliation, religion, and nationality [41]. Perceiving oneself in one specific group fosters in-group favoritism, where individuals give preferential treatment and exhibit positive attitudes toward members with similar traits [42, 43]. When the differences of the outgroup are perceived as nonnormative and inferior, it often results in devaluation and discrimination [44]. Our study examines the parallels between human intergroup biases and the humanlike behaviors exhibited by LLMs.

3 Methods

The goal of this study is threefold. First, we aim to identify political bias in LLMs. Second, we seek to analyze how these models recalibrate their perspectives when prompted to adopt a Republican or Democratic identity as the ingroup. Existing literature suggests that such role-playing would make LLMs more positive and assimilated toward